

1. Small Inter-frame Displacement Assumption

- Object thoda sa hi move kar sakta hai between frames
- **Problem:** Fast-moving objects fail ho jate hain
- Search window ke basin of attraction mein rehna chahiye
- Kadam et al. (2024): "mean shift has problems with tracking fast objects"

2. No Occlusion Handling

- Object hide hua → tracker haar maan jata hai
- Background pe latch kar sakta hai (wrong object)
- Reappear hone pe recover nahi kar pata

3. Appearance Changes ke liye Vulnerable

- Lighting change → drift kar sakta hai
- Static appearance model hai
- Target color = background color → FAIL
- Quote: "cannot significantly distinguish color-similar targets and backgrounds"

4. Template Drift Risk

- Adaptive template updates se background features aa sakte hain
- Gradually target lose ho sakta hai

Result:

Modern benchmarks mein **isolation mein rarely use hota hai**. Hybrid frameworks mein better!

SECTION 3: KALMAN FILTER

3.1 Algorithm Design

Core Concept:

Recursive state estimator jo motion model aur measurements use karta hai.

State Vector (kya track karte hain?):

State = $[x, y, v_x, v_y]^T$